THE COMPLETION OF AN ALGEBRAIC SPACE ALONG A CLOSED SUBSET IS A FORMAL ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the functor whose being a formal algebraic space is asserted; it lies inside the counted span and is reproduced verbatim.

Editorial note. The gate reports a published-overlap note (formal-spaces.tex 3094-3229) for this interval; it is recorded in the dedup block below rather than suppressed. The lane has been cleared by the Lead on the point: no public-face duplication and no two live cards with the same proposition.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Algebraic spaces can be covered by etale morphisms from affine schemes. Properties of Spaces, Lemma spaces-properties-lemma-cover-by-union-affines, invoked at proof lines 3195-3197. Supplies the covering used in the reduction.
- Base changes of a representable etale morphism are etale. Spaces, Lemma spaces-lemma-base-change-representable-transformations-property, invoked at proof lines 3198-3201. Shows the pieces of the reduced situation are etale over the functor and cover it.
- The completion of an affine scheme along a closed subset is an affine formal algebraic space. Same chapter, Lemma lemma-completion-affine-is-affine-formal-algebraic-space, invoked at proof lines 3205-3207. Gives the affine case, which is the last step.

Source reference. Original lemma 0AIY: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3149–3149. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02WL: sources/stacks-chow/tags.txt, source lines 3808–3808. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $T \subset |X|$ be a closed subset. Then the functor*

$$(Sch/S)_{fppf} \longrightarrow Sets, \quad U \longmapsto \{f : U \to X \mid f(|U|) \subset T\}$$

is a formal algebraic space.

Proof. Denote F the functor. Let $\{U_i \to U\}$ be an fppf covering. Then $\coprod |U_i| \to |U|$ is surjective. Since X is an fppf sheaf, it follows that F is an fppf sheaf.

Let $\{g_i : X_i \to X\}$ be an étale covering such that X_i is affine for all i, see Properties of Spaces, Lemma 03FX. The morphisms $F \times_X X_i \to F$ are étale (see Spaces, Lemma 02WL) and the map $\coprod F \times_X X_i \to F$ is a surjection of sheaves. Thus it suffices to prove that $F \times_X X_i$ is an affine formal algebraic space. A U-valued point of $F \times_X X_i$ is a morphism $U \to X_i$ whose image is contained in the closed subset $g_i^{-1}(T) \subset |X_i|$. Thus this follows from Lemma 0AIY. □